

2 INTRODUCTION

2.1 Summary

This chapter describes the terms specifically used in this instruction manual, each head condition, components of the device, and operation outline.

Note that this instruction manual is composed based on the device equipped with the software of the following program numbers:

- DMU:PROGRAM No. T2341E
- RCU:PROGRAM No. W0212C

<Contents>

- Usage, features, and specifications of the device
- Descriptions of the terms used in this manual
- Components of the device and operation outline

<Purpose>

- To understand the basic components, outline, usage, application range, and application limit of the device.

<Intended reader>

- All users

2.2 Intended Use of the Device

This device is intended to weigh and adjust product weight to a set value through computer-controlled combinational weighing.

The device can be used as a weighing instrument for a wide range of products from food such as snacks and candies to machine parts. *1

*1 Application should be limited to products approved by Ishida Co., Ltd. for design reasons.

2.3 Features of the Device

This device is intended to weigh single products.

2.4 Specifications of the Device

This section describes the specifications of the device. Understand the details of this section thoroughly before operating the device.

2.4.1 Standard Specifications

For the specifications of your own device, refer to the separately attached specifications.

For standard specifications, refer to the product catalogue.

2.4.2 Outside Dimension

For the outside dimensions of your own device, refer to the separately attached drawing.

Design and outside dimensions are subject to change without notice.

2.4.3 Operation Environment

Use the device within the operation environment below in order to operate the device effectively and maintain precision as well as to perform weighing safely:

- Indoors
- Ambient temperature between 0 and 40°C
- Ambient relative humidity between 35% and 85% (No condensation allowed)
- Steady location without floor vibration, where the device can be horizontally stable
- Sufficient maintenance space around the device
- No source of radio interference or radio equipment in the vicinity

2.4.4 Audible Noise

The standard machine makes the noise less than 79.6 dBA.

(at a distance of 1 m, a height of 1.2 m to 1.5m, background noise: 41.4 dBA)

Depending on the specification of your own device, this noise level may differ from the above.

Wear earplugs while the machine is in operation, if needed.

2.5 Term Description

The following specific terms are used in this instruction manual.

Table 2-1

Term	Description
Products	The products weighed with this device.
Infeeder	A front-end device that supplies products to this device.
Packer	A post-process device that packages products after being weighed by this device.
Section	The name of the divided head for selecting active heads.
Span	To set the span adjustment weight as a standard weight after zero adjustment.
Drain	To drain products remaining in the device without weighing.
Weighed value	Weight that is combinational weighed.
Target weight	The target weighed value set for each preset.
Upper weight limit	An upper permissible limit, which is relative to the target weight.
Lower weight limit	A lower permissible limit, which is relative to the target weight.
Head	A collective term for the radial trough, pool hopper, weigh hopper and weigh drive unit.
Infeed control	A function to turn ON/OFF the feed signals to the infeder.
Interlock signal (IS)	A product discharging request signal transmitted from the packer to this device.
Overweight	To exceed the upper limit set for the weighed value.
Underweight	To be below the lower limit set for the weighed value.
Proper weight	To be between the upper and lower limits set for the weighed value.
Discharge completion signal (DS)	A signal to inform the outside that the product is discharged.
Overscale	The condition where a product with the weight of (set value + upper limit) or more is supplied to the weigh hopper, or where the weight within the weigh hopper exceeds the head weighing range.
Stagger	A discharge method where discharge timing is staggered and divided into groups in order to prevent the product from jamming at the discharge chute and packer slot when the products are discharged from selected weigh hoppers all at once.
Master mode	A mode in which, when this device is interlocked with the packer, it discharges upon receiving a (continuous) signal from the packer.
Slave mode	A mode in which, when this device is interlocked with the packer, it discharges upon receiving a one shot signal from the packer.
Stroke On Demand mode	A mode in which, when this device is interlocked with the packer, it discharges by transmitting a weighing completion signal to the packer, while in response the packer transmits an interlock signal.

Table 2-1

Term	Description
Bag On Demand mode	A mode in which, when this device is interlocked with the packer, self-calculation is performed until the device discharges a proper amount of product based on a signal from the packer, and a discharge completion signal is transmitted to the packer.
AFV	Stands for Anti Floor Vibration mechanism. With this function, even in a floor vibration environment, the AFV cell detects vibrations, and errors caused by the vibration can be canceled from the load cell output for weighing.
Range	The maximum weight that can be weighed.
Auto zero adjustment	To automatically adjust weight during operations after selecting a hopper with less operation frequency among the combinational selected weigh hoppers, so that the weight display indicates zero.
Preset	Product data on weighing, registered in advance.
Combination	A calculation method where small-divided, several weighed values are combined, and the most favorable combination is used as a weighed value to be discharged.
Full scale	A condition where the weighing sensor value exceeds a permissible load due to external factors.
Multiple dump weighing	A weighing method where the target weight is obtained through multiple dumps, when a large amount of product exceeding the weighing range is weighed or when a product overflows from the weigh hopper if weighed all at once.

2.6 Components of the Device

This section describes the component names and functions of the device.

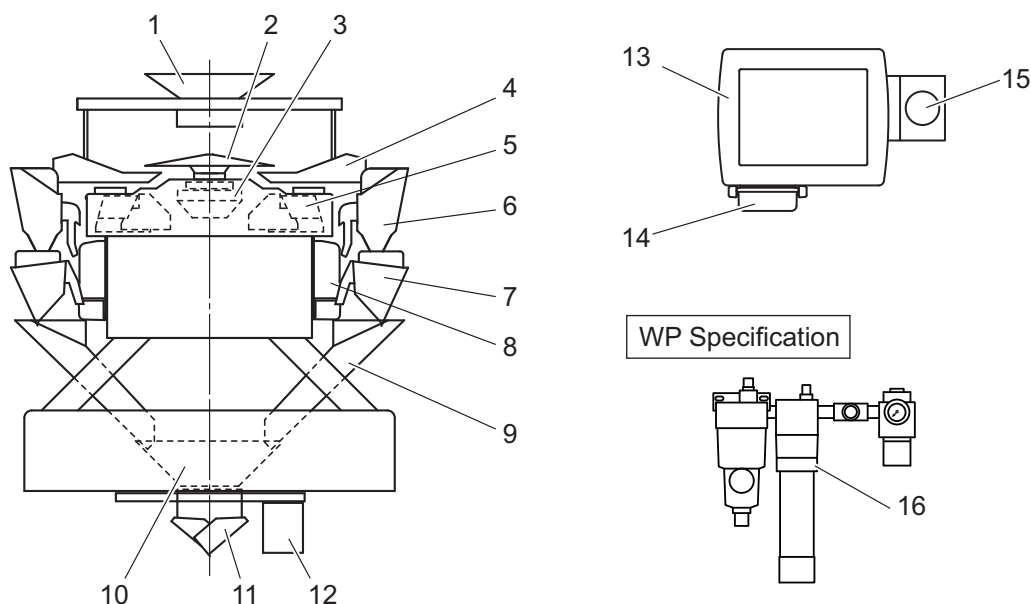


Fig. 2-1

Table 2-2

No.	Name	Function and Description
1	Inlet chute (Optional)	A product feeding inlet. This collects the products supplied from the infeder to the center of the dispersion table.
2	Dispersion table	Supplies products from the inlet chute to the radial troughs.
3	Dispersion feeder (DF)	Vibrates the dispersion table. It is interlocked with the weight sensor and controls the supply by weighing products on the dispersion table.
4	Radial trough	Supplies products from the dispersion table to the pool hoppers.
5	Radial feeder (RF)	Vibrates the radial troughs.
6	Pool hopper (PH)	Temporarily pools products supplied from the radial troughs. It opens and closes in accordance with the weighing operation and supplies products into the weigh hoppers.
7	Weigh hopper (WH)	Weighs supplied products. It opens and closes in accordance with the combinational weighing result and supplies products into the collection chutes.
8	Weigh drive unit (WDU)	Activates open/close of the pool hopper and weigh hopper. It weighs products in the weigh hopper.
9	Collection chute	Collects products discharged from the weigh hopper and supplies them to the discharge chute.
10	Discharge chute	Collects products supplied from the collection chute and supplies them to the following processing device such as the timing hopper and packer.

Table 2-2

No.	Name	Function and Description
11	Timing hopper (TH) (Optional)	Temporarily pools products passed through the discharge chute and discharges them in accordance with the request from the following processing device such as the packer.
12	Timing hopper drive unit (Optional)	Activates open/close of the timing hopper.
13	Remote control unit (RCU)	Performs operation and required setting of the device.
14	Printer	Prints out the total information, including the weighing result.
15	Main power switch	Turns on/off the power supply to the device.
16	Air Dryer	Supplies dry air to the inside of the main body.

2.7 Operation Outline

This section describes the operation outline from weighing and combining the product supplied to the inlet chute to discharging from the timing hopper.

Product flow outline

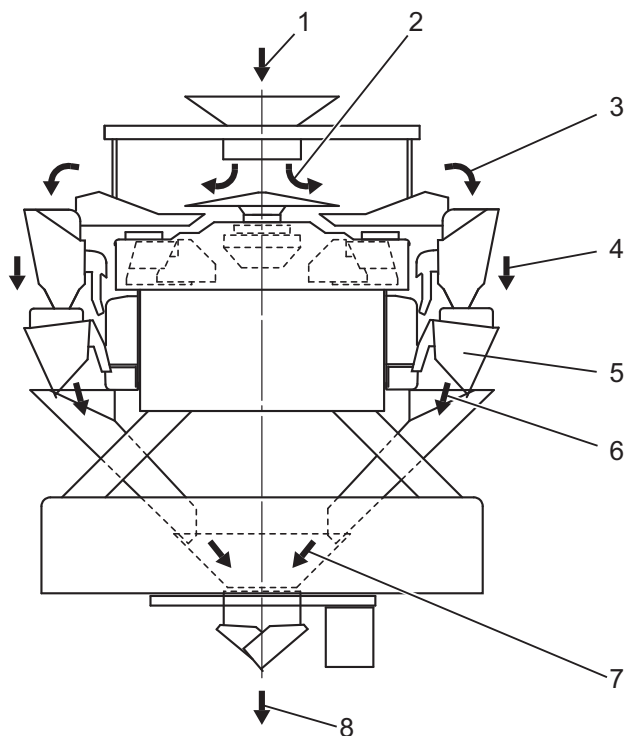


Fig. 2-2

1. The product from the infeeder is supplied to the inlet chute.
2. The product from the inlet chute is supplied to the center of the dispersion table and distributed to the radial troughs.
3. The product from the radial troughs is supplied to the pool hopper.
4. In accordance with the request from the weigh hopper, the product is supplied to the weigh hopper.
5. The product is weighed in the weigh hopper.
6. In order to maintain the closest value to the set weight while preventing underweight, the calculation unit selects an optimized combination of the weigh hoppers, and the products are discharged.
7. The product is collected at the collection chute and discharged from the discharge chute.
8. When the timing hopper is installed, the timing hopper opens and discharges the product in accordance with the discharge request from the packer.